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Experimental and numerical study of hot-rolled duplex stainless steel CHS columns

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Abstract: Stainless steel circular hollow section (CHS) members have been widely used in building structures because of their comparable strength to structural carbon steels, attractive appearance and excellent corrosion resistance. However, research on stainless steel CHS members has been limited to cold-formed members, whereas hot-rolled CHS columns have received less attention. To this end, the flexural buckling performance of hot-rolled duplex stainless steel CHS columns was investigated by a comprehensive experimental and numerical programme. A total of seven hot-rolled duplex stainless steel CHS columns covering two cross-sectional sizes with varying lengths have been tested under compression. Based on the test results, finite element (FE) models were developed and used in parametric studies to generate data points with a wider range of geometric dimensions. Comparing the test and FE results with current design guidance, it was found that the Chinese standard, CECS 410, and North American design guidance, AISC DG 27, underestimate significantly the flexural buckling resistance of such columns, while the design method in Eurocode 3: Part 1-4 might be unsafe in predicting the resistances for low slenderness values. New column buckling curves based on the Eurocode 3 and AISC DG 27 design formulae with improved accuracy and reduced scatters were also proposed and validated through careful reliability analysis.

Keywords: circular hollow section; stainless steel; hot-rolled; design method; flexural buckling

1. Introduction

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23 Stainless steels have the advantages of comparable strength to structural carbon steels, excellent
24 corrosion resistance, low life cycle cost and attractive appearance, all contributing towards their wide
25 applicability in building structures [1]. In recent years, plenty of research work has been done on the
26 flexural buckling performance of stainless steel columns, and fruitful research findings have been
27 achieved. These include Ashraf et al. [2], Afshan et al.[3], Theofanous et al.[4,5], Huang and Young
28 [6-8], Buchanan et al. [9], and He et al. [10], the results of which covered a range of cold-formed
29 hollow sections (circular hollow section (CHS), oval hollow section (OHS), rectangular and square
30 hollow sections (RHS and SHS)) and three material grades (austenitic, ferritic, and duplex). Based
31 on the obtained experimental and numerical results, various existing design methods for cold-formed
32 stainless steel structures have been evaluated and new design recommendations have been proposed.
33 Research into stainless steel welded section members includes Yuan et al. [11,12], Yang et al. [13,14],
34 Shameem et al. [15], Sun et al. [16], Gardner et al. [17] and Bu and Gardner [18], where test and FE
35 studies on austenitic and duplex stainless steel built-up box-section and I-section columns were
36 performed. Recently, Arthur et al. [19] carried out a comprehensive test and FE study on stainless
37 steel hot-rolled angle section columns under axial compression, and updated the corresponding
38 column buckling curves in EN 1993-1-4+A1 [20].

39 Among all the available research works on stainless steel members, the research on CHS
40 columns is mainly limited to cold-formed members [2, 8, 9] and very little information can be found
41 for the hot-rolled stainless steel CHS members. Compared to hot-rolled sections, cold-formed
42 structures generally have smaller wall thicknesses and limited cross-section sizes, hence possessing
43 limited load-carrying capacities. In this regard, hot-rolled members with relatively large wall
44 thicknesses and cross-sectional areas may offer a better structural solution when higher load-carrying

capacities are required in design. The design of hot-rolled stainless steel CHS members refers to various design rules as offered by different design standards in the world [20-22]. The rationality of these design rules still needs to be verified by a sufficient amount of data. In particular, in the Chinese standard, CECS 410 [21], there is no buckling curve specified for the design of hot-rolled stainless steel CHS columns due to the lack of relevant test data.

To this end, a comprehensive experimental and numerical programme on the flexural buckling performance of hot-rolled duplex (grade 2205) stainless steel CHS columns has been performed and is reported in this paper. Seven column tests were conducted, and the test results were used to develop valid FE models. Then the valid FE models were employed for parametric studies to generate more data points with a wider range of geometric dimensions. According to the obtained experimental and numerical results, existing design methods in European standard, EN 1993-1-4+A1 [20], Chinese standard, CECS 410 [21], and North American design guidance, AISC DG 27 [22], were assessed. In addition, new flexural buckling curves, developed based on the current design formulations in these standards [20-22], were proposed to give more accurate predictions to the hot-rolled duplex stainless steel CHS columns.

2. Experimental programme

2.1 Specimen fabrication and material properties

A total of seven hot-rolled duplex (grade 2205) stainless steel CHS columns covering two cross-sectional sizes (114×6 and 219×6) with varying lengths were tested. The measured geometries of the specimens are shown in Table 1, where D is the average outer diameter, t is the average wall thickness of the cross-section, respectively, as denoted in Fig. 1, L represents the cut length of the

column, L_t represents the effective length taking into account of the height of the two knife edges ($L_t = L + 340$ mm) and $\bar{\lambda}$ represents the non-dimensional slenderness of specimen. In the specimen label, 114×6 and 219×6 stand for the nominal cross-section sizes and the last number represents the nominal cut length (in mm) of the specimen.

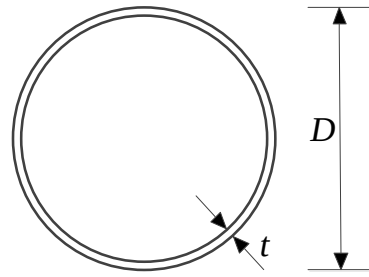


Fig.1 Geometric symbols of hot-rolled CHS

Table 1 Measured geometric dimensions

Specimens	D (mm)	t (mm)	L (mm)	L_t (mm)	$\bar{\lambda}$	Section class
$114 \times 6-3000$	113.9	5.98	3022	3362	1.51	1
$114 \times 6-3500$	114.9	6.44	3522	3862	1.73	1
$114 \times 6-4000$	114.5	6.28	4022	4362	1.96	1
$114 \times 6-4500$	114.4	6.09	4523	4863	2.18	1
$219 \times 6-2500$	219.7	6.34	2524	2864	0.63	3
$219 \times 6-4000$	220.5	7.08	4023	4363	0.96	3
$219 \times 6-4500$	220.3	6.88	4530	4870	1.07	3

The coupons for tensile material tests were cut directly from the reserved tubes that are in the same batch as the test columns. From each cross-section size, three tensile coupons (A, B and C) were extracted from the locations as indicated in Fig 2, resulting in a total of six tensile coupons. The tensile coupon tests were carried out on a Zwick/Roell Z100 loading jack with a maximum tonnage of 10t, following the procedure specified in GB/T 228.1-2010 [23]. The obtained stress-strain (σ - ϵ) curves from the tests are plotted in Fig. 3, and the important material properties, such as 0.2% proof stress ($f_{0.2}$), the Young's modulus (E), the ultimate tensile strength (f_u), the strain hardening exponent (n)

79 and the plastic strain at fracture (ϵ_f), are reported in Table 2

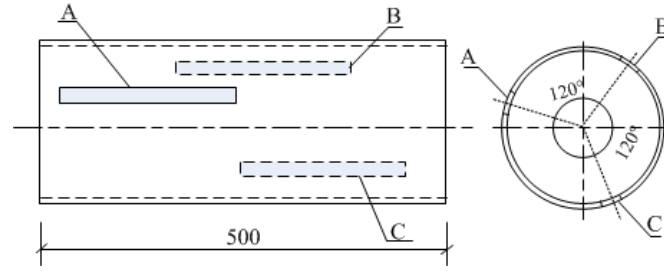


Fig. 2 The sampling locations of the tensile coupons

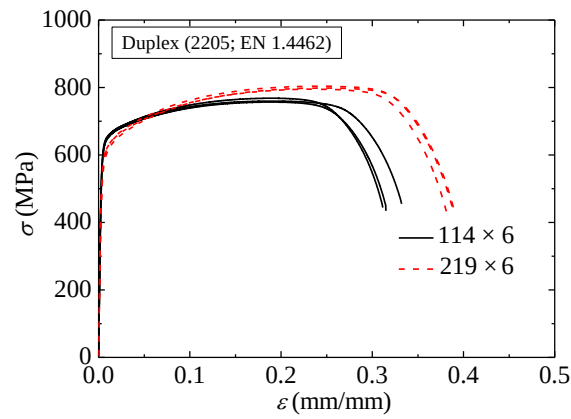


Fig. 3 Measured full stress-strain curves

Table 2 Average material properties obtained from tensile coupon tests

Coupons	E (MPa)	$f_{0.2}$ (MPa)	f_u (MPa)	ϵ_f (%)	n
114×6	207421	606.0	766.0	32.0	5.8
219×6	205678	560.0	800.0	38.7	4.3

83

84 2.2 Geometric imperfections

85 Since all test columns were designed to be susceptible to overall flexural buckling, only the
86 global imperfections of the columns were measured. The measurement adopted the pull wire method,
87 in which a thin wire was drawn longitudinally between the two ends of the test column and was
88 pressed as closer as possible to the surfaces of the column ends. The thin wire is pressed as much as
89 possible to fit the outer wall of the end of the column. The deviations of the wire to the tube surface
90 at the quartering points were measured for four times and the maximum value was taken as the

imperfection values (δ_1 , δ_2 and δ_3) at these points, as shown in Fig.4. The maximum value of δ_1 , δ_2 and δ_3 was reported as the global geometric imperfection (δ_0) of the specimen, as summarised in Table 3. The loading eccentricity (e) of each specimen can be determined by $e = (e_b + e_t)/2$ [24], where e_b and e_t are the loading eccentricity of the top and bottom of the specimen respectively. The equivalent initial eccentricity (e_{Eq}) approximately equal to e plus δ_0 , is also reported in Table 3 for each specimen. It can be seen that the e_{Eq} values for all specimens are relatively small (below $L/1000$).

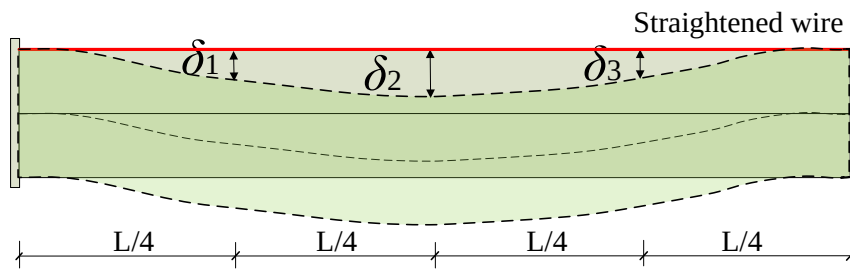


Fig. 4 Global initial imperfection measurement

Table 3 Measured initial geometric imperfections

Specimens	Initial geometric imperfections (mm)				Loading eccentricity(mm)		e_{Eq}/L
	δ_1	δ_2	δ_3	δ_0	e_b	e_t	
114 × 6-3000	1.12	1.28	0.78	1.28	-0.081	0.270	1/2183
114 × 6-3500	0.99	0.43	1.78	1.78	-0.028	0.265	1/1844
114 × 6-4000	2.00	1.65	1.70	2.00	-0.068	0.324	1/1880
114 × 6-4500	2.00	0.87	1.32	2.00	-0.213	0.104	1/2313
219 × 6-2500	0.34	0.97	1.50	1.50	-0.370	-0.457	1/2301
219 × 6-4000	0.45	1.08	1.40	1.40	-0.286	0.560	1/2602
219 × 6-4500	1.55	2.56	1.27	2.56	-0.247	0.729	1/1607

98

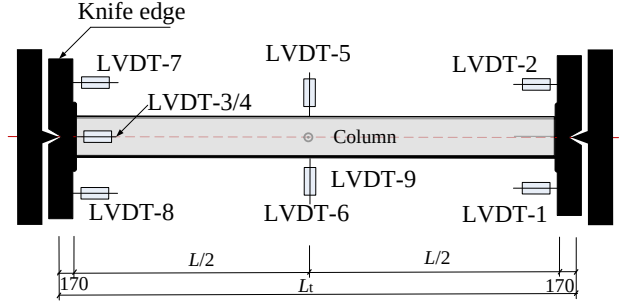
99 2.3 Flexural buckling tests

100 2.3.1 Loading setup and measurement arrangement

101 The WAW-2500 hydraulic universal testing machine, as shown in Fig. 5(a), was used to load the
 102 specimens. The specimens were pinned supported at the two ends using a pair of knife edges, which
 103 allow the rotation around the buckling axis.



(a) Loading setup



(b) Instrumentation scheme

Fig. 5 Test configuration

During the tests, the load, deflection and strain development of the specimens were recorded. The axial deformation, lateral displacement and end rotations of the columns were measured by nine linear variable displacement transducers (LVDTs), as depicted in Fig. 5(b). LVDT-1, 2, 7 and 8 were used to monitor the in-plane rotations of the column ends, LVDT-3 and 4 were located at the moving end of the loading jack to measure the axial deformation of the column, LVDT-5 and 6 were located at the middle height of the column to obtain the in-plane lateral displacement, and LVDT-9 was placed at the middle height cross-section to monitor the out-of-plane deflection. The load was read directly from the loading jack. The strain development during the tests was measured by three groups of four strain gauges arranged at the end (30mm away from the end-plate) and the mid-height cross-sections according to the arrangement shown in Fig. 6. The SG-1, 2, 3 and 4 and SG-9, 10, 11 and 12 were placed at the column ends and the SG-5, 6, 7 and 8 were placed at the middle height of the column.

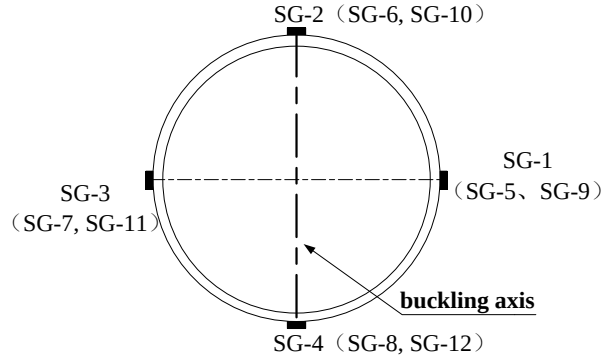


Fig. 6 Layout of strain gauges

2.3.2 Test results

The ultimate bearing capacities, N_u , of all specimens obtained in the tests are reported in Table 4. All the column specimens exhibited overall buckling instability before failure, and the typical failure mode is shown in Fig. 7. Fig. 8 shows the load versus lateral displacement at mid-height ($N-w$) curves of all the specimens. As expected, longer specimens displayed smaller ultimate resistances for the same cross-section sizes. In Fig. 8 (a), piecewise linear plateaus are found in the axial load versus lateral displacement curves for specimens $114 \times 6-4000$ and $114 \times 6-4500$. This is mainly due to the large tonnage of experimental loading equipment, which is not sensitive enough to the change of small force.



Fig. 7. Typical failure mode

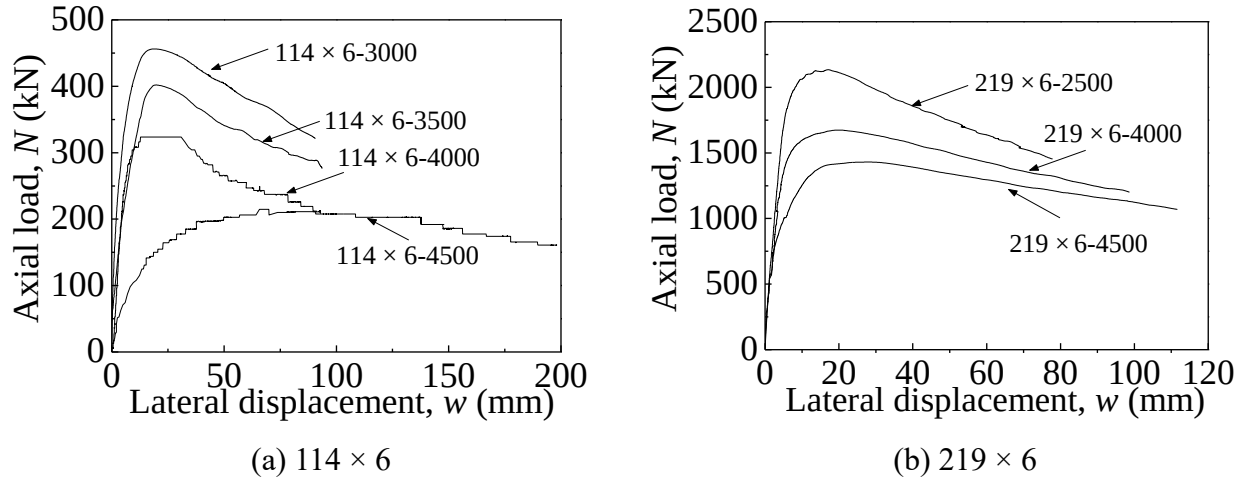


Fig. 8 Axial load versus lateral displacement at mid-height of columns

126 3. Numerical modelling

127 3.1 FE modelling technique

128 Following the experimental research, a numerical programme was performed to extend the test
 129 dataset. The FE software ANSYS was employed in the numerical study. The BEAM188 element,
 130 which has been successfully and extensively utilised in similar study to model tubular members under
 131 axial compression [13], was also adopted in the current study. The FE model and mesh are illustrated
 132 in Fig. 9, where the column was divided into 40 equal parts along the circumference of the section
 133 and 50 (for $L < 4000$ mm) or 100 (for $L > 4000$ mm) segments longitudinally. The boundary conditions
 134 were set as shown in Fig. 9, where the vertical movement of the loading end and the rotation about
 135 the buckling axes at both ends were released, and the other degrees of freedom at both ends were
 136 fixed.

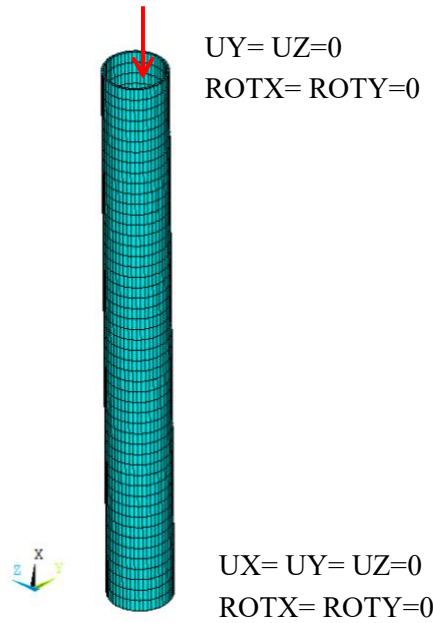


Fig. 9 FE model

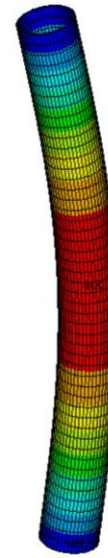


Fig.10 First global eigenmode

137 Based on the measured stress-strain responses (Fig. 3) and the measured material properties
 138 reported in Table 2, a multi-linear kinematic hardening constitutive model was established and
 139 employed in the numerical modelling. Since all the experimental specimens displayed a global
 140 buckling failure mode, only global geometric imperfection was considered in the FE models. The
 141 shape of the first global eigenmode from Eigenvalue analysis (Fig. 10) was used to represent
 142 imperfection pattern, and the previously measured amplitudes of global geometric imperfections, as
 143 shown in Table 3, were input using the ‘UPGEOM’ command. The residual stresses were not
 144 considered in the FE models owing to the relatively low levels of membrane residual stresses as
 145 commonly found in hot-rolled members [25, 26]. The membrane residual stresses of hot-rolled
 146 sections are usually less than 10% of $f_{0.2}$ [27], and it has been proved in reference [28] that relatively
 147 low residual stress level has little effect on the overall performance and bearing capacity of stainless
 148 steel CHS columns. In similar studies on hot-rolled section columns [29, 30], residual stresses were
 149 also not particularly considered in the FE models. Providing the above arguments, the FE models in
 150 the current study did not include the residual stresses.

151 3.2 Validation

152 The FE results are compared with the corresponding test results in terms of the failure modes
153 and the load-displacement relationships to verify the accuracy of the FE models. Fig. 11 shows typical
154 comparisons between the failure modes achieved in the tests and the corresponding FE models, where
155 agreement between the two has been achieved in both cases in Figs 11 (a) and (b). The FE and test
156 load-lateral displacement curves of specimen $114 \times 6-3000$ and $219 \times 6-4000$ are compared in Figs.
157 12 (a) and (b), respectively. These selected examples demonstrate that the FE models can accurately
158 capture the load-displacement responses as experienced by the test specimens.

159 The obtained test and FE ultimate load, $N_{u,\text{test}}$ and N_{FE} , are given in Table 4 for all the specimens.
160 The mean FE-to-test ultimate load ratio $N_{\text{FE}}/N_{u,\text{test}} = 1.00$, confirming the validity of the developed
161 FE models. In general, the developed FE models are shown to produce satisfactory predictions and
162 can be employed in the subsequent parametric research.

163

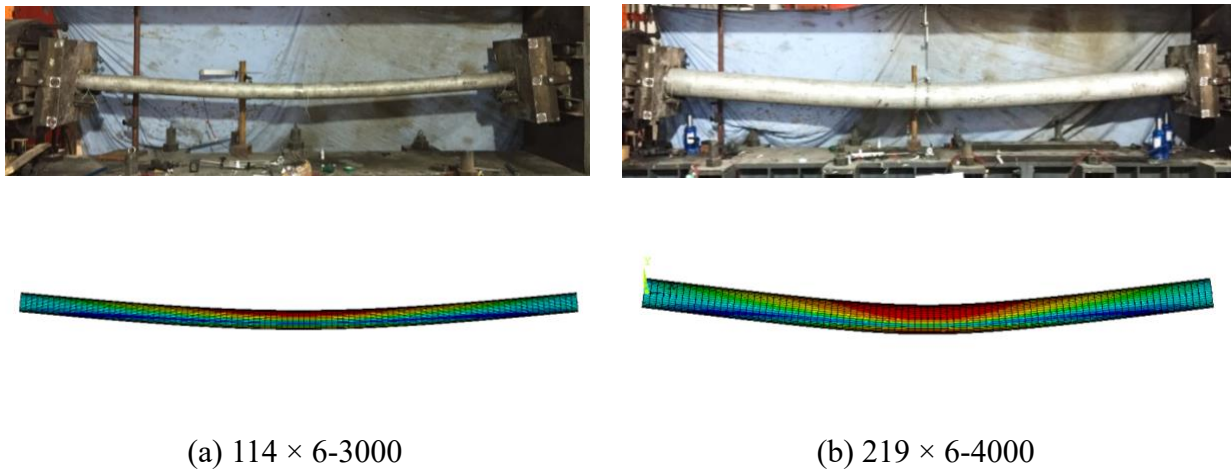


Fig.11 Comparison of test and FE failure modes

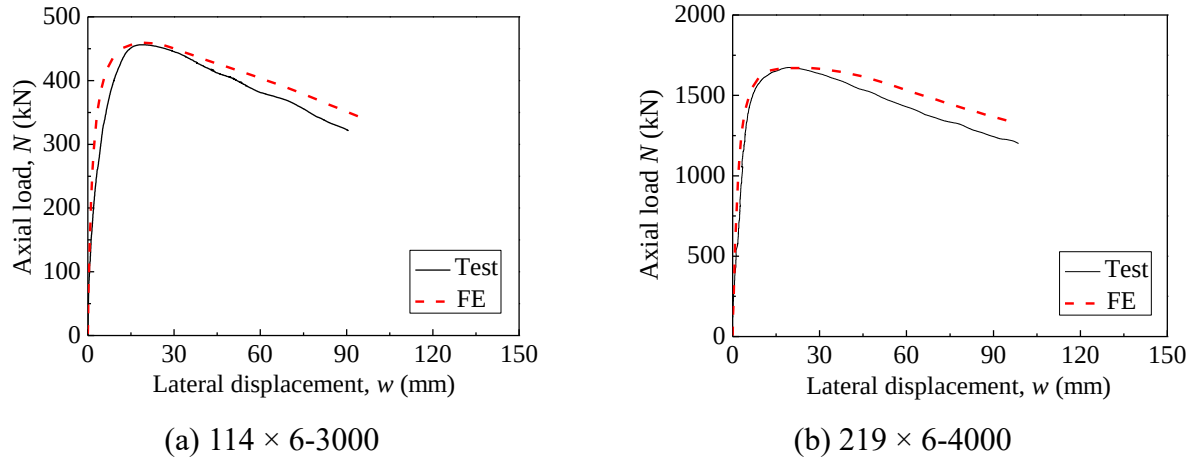


Fig. 12 Comparison between FE and test load-lateral displacement curves

Table 4 Summary of test and FE results

Specimen	$N_{u.test}$ (kN)	N_{FE} (kN)	$N_{FE} / N_{u.test}$
114 × 6-3000	456.8	459.4	1.01
114 × 6-3500	402.0	406.4	1.01
114 × 6-4000	323.6	327.4	1.01
114 × 6-4500	214.8	216.9	1.01
219 × 6-2500	2134.1	2092.4	0.98
219 × 6-4000	1673.6	1670.0	1.01
219 × 6-4500	1430.4	1439.7	1.01
Mean			1.00
COV			0.01

3.3 Parametric studies

Employing the validated FE models, parametric studies were carried out. Table 5 details the key variables in the parametric studies, in which nine cross-sections, covering three cross-sectional classes (Class 1, 2 and 3) [20] and three wall thicknesses ($t = 4, 6$ and 8 mm), varying in lengths were considered. In Table 5, $D/t\epsilon^2$ is the cross-sectional slenderness as defined in EN 1993-1-4+A1 [20], being $\epsilon = \sqrt{235E_0/(210000f_{0.2})}$. It should be noted that only fully effective cross-sections (Class 1, 2 and 3) were considered in the parametric studies whereas the partially effective (Class 4) cross-sections which may experience premature local buckling failure are out of the scope of the current study. The input material parameters and stress-strain relationships were defined according to the averaged tensile coupon test results. The amplitude of global geometric imperfection in FE models is taken as $L_t/1000$. In total, there were 115 FE models generated in the parametric studies.

176

Table 5 Summary of specimen geometries in the parametric studies

Grade	Section	D (mm)	t (mm)	D/t	$D/t\epsilon^2$	Class	$\bar{\lambda}$	No
Duplex (2205; EN 1.4462)	64×4	64	4	16.0	40.4	1	0.20~2.0	11
	95×4	95	4	23.8	59.9	2	0.20~2.0	11
	127×4	127	4	31.8	80.1	3	0.20~2.0	11
	95×6	95	6	15.8	39.9	1	0.10~2.3	27
	152×6	152	6	25.3	63.9	2	0.20~2.0	11
	180×6	180	6	30.0	75.7	3	0.20~2.0	11
	114×8	114	8	14.3	35.9	1	0.20~2.0	11
	180×8	180	8	22.5	56.8	2	0.20~2.0	11
	245×8	245	8	30.6	77.2	3	0.20~2.0	11

177

4. Discussion and assessment of current design methods

178

Three design methods for hot-rolled duplex stainless steel CHS axial compression members, as

179

included in the European standard, EN 1993-1-4+A1 [20], Chinese standard, CECS 410 [21], and

180

North American design guidance, AISC DG 27 [22], were introduced and evaluated by means of the

181

experimental and numerical results. Based on the assessment, modified design recommendations

182

were proposed in Section 5.

183

4.1 EN 1993-1-4+A1

184

The design method for axial compression members in EN 1993-1-4+A1 is derived from the Perry

185

Robertson buckling formula [31]. The resistance to flexural buckling ($N_{b,Rd}$) of members under axial

186

compression is calculated according to Eqs (1-4) for Class 1, 2 and 3 cross-sections.

$$N_{b,Rd} = \frac{\chi A f_{0.2}}{\gamma_{M1}} \quad (1)$$

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \leq 1 \quad (2)$$

$$\phi = 0.5 \left(1 + \alpha (\bar{\lambda} - \bar{\lambda}_0) + \bar{\lambda}^2 \right) \quad (3)$$

$$\bar{\lambda} = \sqrt{\frac{A f_{0.2}}{N_{cr}}} \quad (4)$$

187 where A is the cross-sectional area, γ_{M1} is the partial safety factor which is set to unity in the current
 188 study, and χ is the buckling reduction factor calculated according to Eqs (2-3) as a function of $\bar{\lambda}$ (as
 189 defined in Eq. (4), being N_{cr} the Euler buckling load). In EN 1993-1-4+A1, an imperfection factor of
 190 $\alpha = 0.49$ and a limiting slenderness of $\bar{\lambda}_0 = 0.4$ are adopted in the buckling curve for stainless steel
 191 hollow sections (welded and seamless).

192 4.2 CECS 410

193 In CECS 410, the design equation adopts the same format as in EN 1993-1-4+A1 (Eqs (1-4))
 194 except that different values of α and $\bar{\lambda}_0$ are used. In CECS 410, six buckling curves are given
 195 considering different fabrication processes and cross-section shapes (two for cold-formed and four
 196 for welded section members). The buckling curve of hot-rolled stainless steel CHS columns was not
 197 specifically defined in CECS 410 because of the lack of test data of such column. Therefore, for hot-
 198 rolled duplex stainless steel CHS columns, CECS 410 only gives conservative design
 199 recommendations (α and $\bar{\lambda}_0$ are conservatively taken as 0.89 and 0.26, respectively.)

200 4.3 AISC DG 27

201 In AISC DG 27, a single column buckling curve is adopted to design stainless steel non-slender
 202 element compression members, as expressed in Eqs. (5-7), where A_g is the gross area of the cross-
 203 section and the strength reduction due to flexural buckling is considered in the calculation of the
 204 critical stress F_{cr} .

$$N_{b,Rd} = F_{cr} A_g \quad (5)$$

$$F_{cr} = \left(0.50^{\frac{F_y}{F_e}} \right) F_y \quad \text{for} \quad \frac{F_y}{F_e} \leq 1.44 \quad (6)$$

$$F_{cr} = 0.531 F_e \quad \text{for} \quad \frac{F_y}{F_e} > 1.44 \quad (7)$$

205 4.4 Discussion and assessment

206 The design approaches for hot-rolled duplex stainless steel CHS axial compression members in
 207 the above three design standards were evaluated based on the test and FE results. Fig. 13 presents the
 208 three buckling curves with test and FE data, which are plotted in terms of the global slenderness $\bar{\lambda}$
 209 versus the normalised resistance $N_u/Af_{0.2}$ (equivalent to buckling reduction factor). It can be seen from
 210 Fig.13 that for such columns, the EN 1993-1-4+A1 buckling curve provides more accurate strength
 211 predictions than the CECS 410 and AISC DG 27 curves, but is unsuitable for low slenderness values
 212 with overestimated predictions of the ultimate capacity. The CECS 410 curve underestimates
 213 significantly the resistances in the intermediate slenderness range ($\bar{\lambda} = 0.5$ to 1.5) and the AISC DG
 214 27 curve is increasingly conservative in the high slenderness range.

215 Numerical comparisons, in terms of the ratio between the obtained ultimate capacity, N_u , and the
 216 prediction, $N_{u,EC3}$ ($N_{u,CECS}$ and $N_{u,AISC}$), are summarised in Table 6. The mean value of $N_u/N_{u,EC3}$ equals
 217 to 1.04 and COV of 0.05, which confirms again the conclusion that the method in EN 1993-1-4+A1
 218 provides the closest strength prediction among the three. The American approach gives the most
 219 conservative estimation of column capacities with the highest $N_u/N_{u,AISC}$ ratios (1.27) and COV
 220 values (0.18).

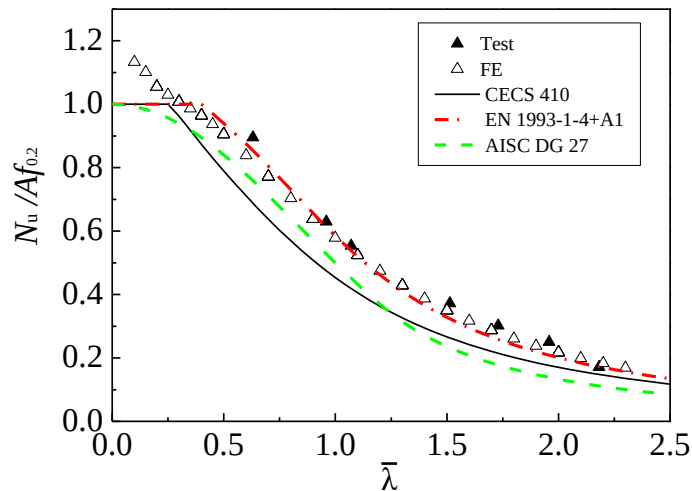


Fig. 13 Comparison of the test and FE results with existing buckling curves

Table 6 Comparison of test and FE results with predicted strengths $N_u / N_{u,pred}$

No. of tests	No. of FE models	$N_u / N_{u,EC3}$		$N_u / N_{u,CECS}$		$N_u / N_{u,AISC}$		$N_u / N_{u,pro,1}$		$N_u / N_{u,pro,2}$	
		Mean	COV	Mean	COV	Mean	COV	Mean	COV	Mean	COV
7	115	1.04	0.05	1.22	0.08	1.27	0.18	1.04	0.03	1.09	0.09

5. New flexural buckling curve proposals

The assessment in Section 4 indicates that the three design approaches generally provide overestimations of the bearing capacities of hot-rolled duplex stainless steel CHS axial compression members. In order to improve their accuracy, modified buckling curves developed based on the obtained test and FE data were proposed and are introduced in this section.

5.1 Modified buckling curves for EN 1993-1-4+A1 and CECS 410

Both of the flexural buckling expressions in CECS 410 and EN 1993-1-4+A1 adopt the Perry Robertson formula but with different imperfection factors and limiting slendernesses. Modification to these design equations is essentially the proposal of suitable imperfection factors and limiting slendernesses. Calibrated against the test and FE data, it is proposed to adopt $\alpha = 0.42$ and $\bar{\lambda}_0 = 0.26$ for the design of hot-rolled duplex stainless steel CHS columns

The proposal (named as Proposal 1 herein) is compared with the test and FE results and the corresponding EN 1993-1-4+A1 and CECS 410 design curves in Fig. 14. In general, the proposed curves provide a closer fit to the test and FE data than the codified curves while maintaining the conservativeness. The normalised test and FE capacities $N_u / N_{u,pred}$ by the CECS 410 and EN 1993-1-4+A1 predictions varying with $\bar{\lambda}$ are given in Fig. 15, where the results normalised by the Proposal 1 predictions are also included for comparison purpose. It can be seen in Fig. 15 that the Proposal 1

239 gives more uniform estimation of the test and FE results with reduced scatters for all the length
 240 domains than the codified methods. The numerical values of $N_u / N_{u,pred}$ plotted in Fig. 15 are also
 241 summarised in Table 6. The mean value of $N_u / N_{u,pro,1}$ equals to 1.04, which is superior to the
 242 corresponding CECS 410 predictions. The coefficients of variations (COV) of $N_u / N_{u,pro,1}$ equals to
 243 0.03, which is superior to the corresponding CECS 410 and EN 1993-1-4+A1 predictions. Therefore,
 244 it can be concluded that the Proposal 1 can provide more satisfactory prediction to the resistance of
 245 such columns than those given in EN 1993-1-4+A1 and CECS 410.

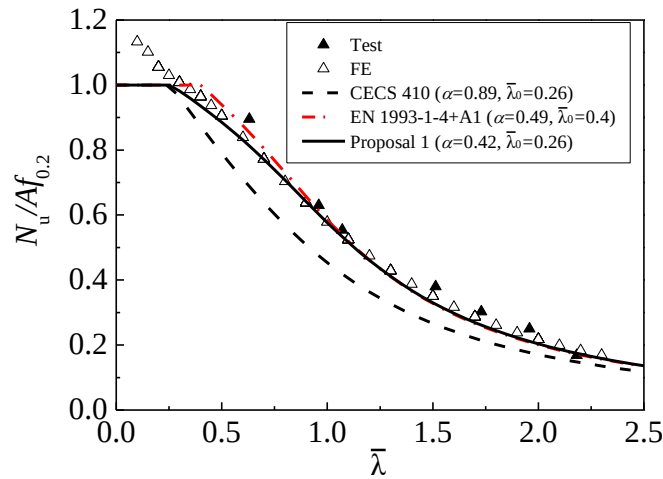


Fig. 14 Comparison of the CECS 410, EN 1993-1-4+A1 and Proposal 1 buckling curves with the test and FE data

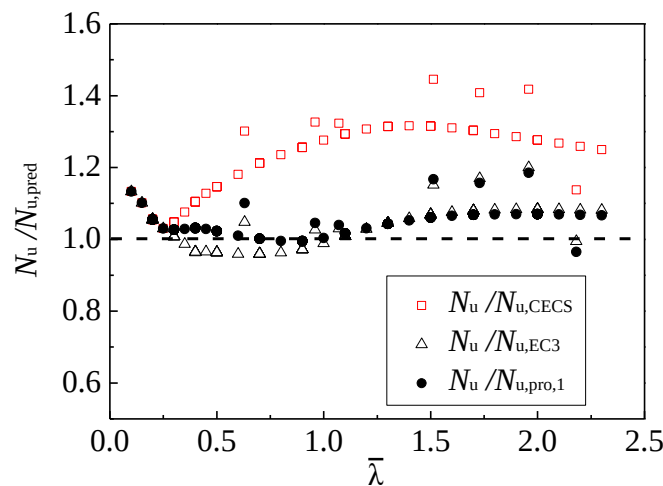


Fig. 15 Test and FE data normalised by the predictions made by the proposed design curve (Proposal 1), the CECS 410 and EN 1993-1-4+A1 design curve varying against the column slenderness

246 5.2 New flexural buckling curve proposals for AISC DG 27

247 The flexural buckling curve expression in AISC DG 27 (Eqs (6) and (7)) is defined as a two-
 248 stage curve and can be rewritten into the format of Eqs (8) and (9) where a_1 is the factor that defines
 249 the division of the two stages, a_2 is the factor that governs the buckling resistance in the inelastic
 250 range and a_3 is the reduction factor to the Euler resistance in the elastic range.

$$F_{cr} = \left(a_2^{\frac{F_y}{F_e}} \right) F_y \quad \text{for } \frac{KL}{r} \leq a_1 \sqrt{\frac{E}{F_y}} \quad (8)$$

$$F_{cr} = a_3 F_e \quad \text{for } \frac{KL}{r} > a_1 \sqrt{\frac{E}{F_y}} \quad (9)$$

251 It should be noted that this format of expression is also adopted in ANSI/AISC 360-10 [32] for
 252 the design of conventional steel structures, and AISC DG 27 retains the same format but adopts
 253 smaller values of a_1 , a_2 and a_3 for the design of stainless steel members. In order to give a better
 254 prediction to the resistances of hot-rolled duplex stainless steel CHS columns, updated values of $a_1 =$
 255 4.24, $a_2 = 0.586$ and $a_3 = 0.69$ were derived based on the test and FE data aiming at producing the
 256 closest yet safe fit to the dataset. This proposal is named as Proposal 2 herein.

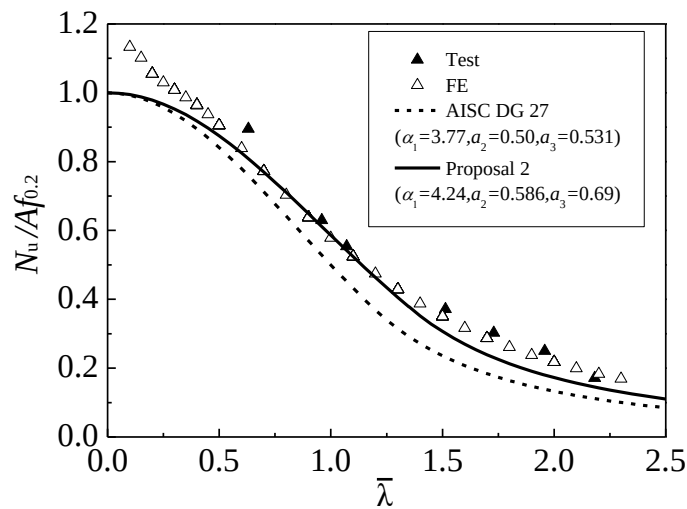


Fig. 16 Comparison of the AISC DG 27 and Proposal 2 buckling curves with the test and FE data

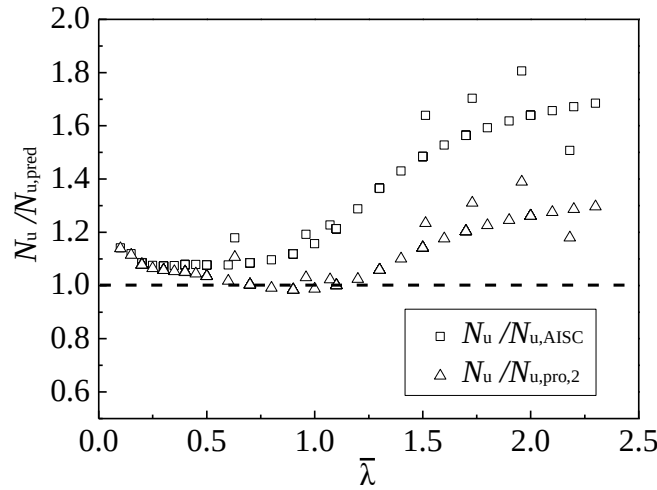


Fig. 17 Test and FE data normalised by the predictions made by the proposed design curve (Proposal 2) and the AISC DG 27 design curve varying against the column slenderness

As shown in Fig. 16, the Proposal 2 is compared with the test and FE results and the AISC DG 27 curve. The ratios of the test and FE results to the AISC DG 27 and Proposal 2 predictions varying with column slenderness are plotted and compared in Fig. 17, and the averaged value of $N_u / N_{u, pro,2}$ is reported in Table 6. It can be seen from both Fig. 17 and Table 6 that the Proposal 2 design curve reports $N_u / N_{u, pro,2}$ values that are much closer to unity with reduced scatters compared to the AISC DG 27 design curve, suggesting that the Proposal 2 is more suitable for the design of such columns than AISC DG 27. This conclusion has also been supported by the results from reliability analysis, as detailed in Section 6.

6. Reliability analysis

In accordance with the guidelines of EN1990-Annex D [33], reliability analysis was carried out to verify the validity of the design methods in the above three design standards and the proposed buckling curves (Proposals 1 and 2) for hot-rolled duplex stainless steel CHS columns. According to the recommendation of reference [34], in the reliability analysis, the material over-strength factor for hot-rolled duplex stainless steel used was 1.10, and the coefficients of variation of the yield strength

271 V_{fy} and geometry $V_{geometry}$ were respectively 0.03 and 0.05. The results of reliability analysis are
 272 summarized in Table 7, where γ_{M1} is partial safety factor, V_{δ} is the estimator for the coefficient of
 273 variation of the error term, b is the average ratio of the test/FE resistance to model resistance, and $k_{d,n}$
 274 is the design fractile factor and is related to the number of test and FE results (n).

275 From Table 7, it can be seen that the EN 1993-1-4+A1 design curve report γ_{M1} value that does
 276 not meet the target of $\gamma_{M1} \leq 1.1$ as set out in EN 1993-1-4+A1 [20], while CECS 410 and Proposal 1
 277 design curves report γ_{M1} values that meet the target of $\gamma_{M1} \leq 1.1$. Among these three curves, the
 278 Proposal 1 curve is able to achieve higher accuracy with reduced scatters in predicting the resistances
 279 of hot-rolled stainless steel CHS columns (Table 6) while satisfying the Eurocode reliability
 280 requirement. The required partial safety factor of AISC DG 27 ($\gamma_{M1} = 1.374$) fails significantly to
 281 meet the target value of $\gamma_{M1} \leq 1.1$. The Proposal 2 buckling curve developed in the framework of the
 282 AISC DG 27 curve reports γ_{M1} value of 1.146 that is marginally beyond the target of 1.1, but it has
 283 significantly improved accuracy compared to the AISC DG 27 design curve, as can be reflected by
 284 the reduced scatters ($V_{\delta} = 0.084$) and b value ($b = 1.05$) that is closer to unity. The relatively higher
 285 γ_{M1} factors required by the AISC DG 27 and Proposal 2 design curves may attributed to their higher
 286 scatters in the resistance predictions (i.e. higher V_{δ} value).

287 **Table 7** Results of the reliability analysis

Design method	n	$k_{d,n}$	b	V_{δ}	γ_{M1}
CECS 410	122	3.17	1.13	0.087	1.088
EN 1993-1-4+A1	122	3.17	1.00	0.052	1.148
AISC DG 27	122	3.17	1.11	0.170	1.374
Proposal 1	122	3.17	1.03	0.033	1.053
Proposal 2	122	3.17	1.05	0.084	1.146

288 7. Conclusions

289 A comprehensive experimental and numerical investigation focusing upon the flexural buckling

performance of hot-rolled duplex stainless steel CHS columns subjected to axial compression is presented herein. The obtained experimental and numerical results have been compared against the existing design methods in European standard, EN 1993-1-4+A1, Chinese standard, CECS 410, and North American design guidance, AISC DG 27. It was found that the design methods in CECS 410 and AISC DG 27 are conservative in predicting the resistances of hot-rolled duplex stainless steel CHS columns, while the design method in EN 1993-1-4 + A1 might be unsafe in predicting the resistances for low slenderness values.

Based on the assessment results of current design methods, new column buckling curves, adopting the format of the flexural buckling expressions in AISC DG 27 and EN 1993-1-4+A1, were proposed. The superiority of the proposed buckling curves over the current codified design methods has been confirmed through careful reliability analyses.

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